

Exploring sustainable pathways for cyanide biodegradation in cassava waste: Insights into methodologies, microbial degradation, and emerging technologies

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Highlights

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Cyanide concentration in cassava waste is variable in the industrial sector.

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Five enzymatic pathways are described in cyanide removal.

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Added-values products can be obtained from cassava waste.

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Cyanide can inhibit the activity of several microorganisms.

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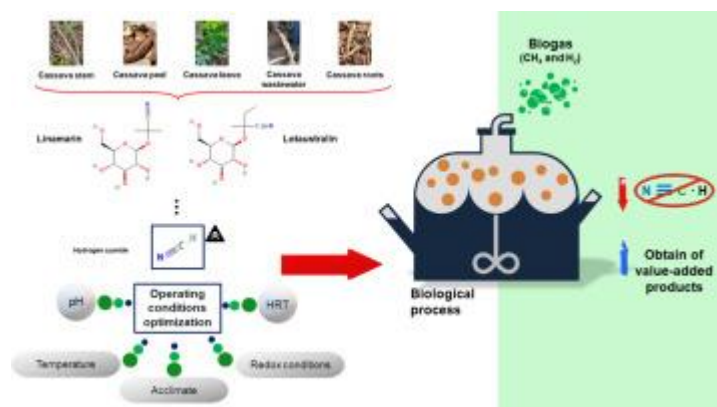
Acclimatization is crucial for efficient cyanide biodegradation.

Abstract

Cassava is a food worldwide used to manufacture various products. From the harvest to the processing of this tuber, substantial amounts of liquid and [solid waste](#) are generated with a significant of organic matter and cyanide (CN). The characteristics of cassava waste are well-documented, but some divergent parameters are due to [climatic factors](#), variety of species, and type of industry. The CN quantification in cassava waste is performed with different methodologies. However, the direct methods based on [ninhydrin](#) are simple, fast, and highly selective. Under anaerobic and [aerobic conditions](#) CN can be degraded. Optimization of operational conditions (e.g., temperature, and pH) and acclimatization are crucial to prevent inhibition of [microorganism](#) and efficiency in CN biodegradation.

Furthermore, anaerobic granular inoculum seems to be a viable alternative to CN removal due to a combination of simultaneous abiotic and biotic processes. In addition, there are microorganisms such as *S. cerevisiae* capable of CN tolerating and degrading. Conventional biological processes are well established for CN removal. However, other alternative processes such as electro-biodegradation, CN-degrading enzyme immobilization, [microbial fuel cell](#) (MFC), and two-step [anaerobic digestion](#) (AD), are prominent due to efficiency and obtaining value-added products (VAP), being considered promising methods for the CN biodegradation.

Graphical abstract



1. [Download: Download high-res image \(166KB\)](#)
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Introduction

Cassava or *manioc* (*Manihot esculenta* Crantz) is a shrub belonging to the *Euphorbiaceae* family, which presents roots with considerable amounts of starch, protein, and other nutrients [[1], [2], [3], [4]]. Therefore, it is an essential crop in more than ninety-nine countries, playing an important socio-economic role in rural areas, especially in developing countries [5,6]. More than half of worldwide cassava is produced by Nigeria, Thailand, Democratic Republic of the Congo, Ghana, Cambodia, Brazil, and Indonesia representing 218×10^6 tons in 2022 [7]. The processing of cassava roots produces a large amount of pollution (e.g., solid, liquid, and gaseous emissions), and their direct disposal can cause environmental problems [[8], [9], [10], [11]]. One ton of processed cassava root can produce from 300 to 600 L of cassava wastewater (CWW) [12,13]. Other authors also report from 250 to 600 kg of CWW produced for each ton of cassava processed [9]. The CWW, also known as *manipueira*, is a yellowish liquid characterized by highly suspended solids (e.g., lipids and non-soluble carbohydrates) and cyanide (CN) concentration, which is a toxic substance that needs the correct waste management [[14], [15], [16]].

The cyanide (CN) is produced by plants, animals, and microorganisms as a defense mechanism [[17], [18], [19], [20]]. As result, CN binds to metals, making metalloenzymes (e.g., cytochrome-c oxidase - Cox), which impaired cell respiration [[21], [22], [23], [24], [25]]. The exposition of humans to $1 \text{ mg} \cdot \text{kg}^{-1}$ per body weight of hydrogen cyanide gas (-HCN) can result in seizures and cardiac or cardiorespiratory arrest. Furthermore, higher doses are lethal for humans [19,[26], [27], [28]]. The CN amount found in cassava differs among species in the range of 0.075 to 1 gCN per kg of root [15]. In CWW, the CN concentrations have been usually reported to range from 26 to 89 $\text{mg} \cdot \text{L}^{-1}$. However, this value can reach $>200 \text{ mg} \cdot \text{L}^{-1}$ depending on the root processing [14,27,29,30]. Therefore, wastewater cannot be released into water bodies without proper treatment [31]. CN is released as hydrogen cyanide (-HCN) when cyanogenic glycosides (CNG) in cassava are hydrolyzed [32,33].

Due to the large volume of waste containing CN generated in the cassava processing industry, it is necessary to lead biological treatment strategies that contribute to the green economy. Some reviews report only CN degradation from mining and steel manufacturing wastewater or detail the CN degradation by enzymatic pathways [[34], [35], [36], [37], [38]]. However, there is no emphasis on the wastewater rich in CN and organic matter, such as cassava waste, and their potential contribution to the biorefinery concept, in addition to critically discussing the CN inhibitory effect on biological processes to obtain products of interest. This review discusses the steps of waste generation in the cassava industry, their physicochemical characteristics, methodologies for CN quantification, the CN removal present in cassava processing waste addressed by biological processes, and optimal operating conditions (pH, temperature, retention time, and C/N ratio). In addition, it was critically discussed how the presence of CN could interfere with obtaining value-added products (VAP) by biological treatment processes. The objective of this report is to point out directions for decision-making regarding CN biodegradation from CWW, concomitantly with the generation of VAP. To the best of our knowledge, this is the first review paper reporting the CN biodegradation from cassava waste by biological processes.

Bibliographic survey

The literature search was carried out in the Scopus database (Fig. 1-A) from 2010 to 2023 with the keywords “biodegradation” OR “treatment” AND “cassava waste”. The search returned the following results within the search area: 320 (75.47 %) are articles, 52 (12.26 %) are conference papers, 27 (6.37 %) are reviews, 15 (3.58 %) are conference reviews, and 10 (2.36 %) are book chapters. According to Fig. 1-A, the period from 2014 to 2016 the number of documents found declined, but in recent years

Cassava transformation process and waste generation

During harvest and transport, cassava is characterized by a significant loss of raw material (around 25 %), making tuber storage the most considerable challenge [41,42]. Cassava solid waste (CSW) is found in different forms: cassava bagasse (CB) or pulp (CP), peel (CPe), stem (CSt), rhizome (CR), leaf (CL), and periderm [10,43,44]. As previously mentioned, the cassava industry generates a high amount of solid and liquid waste, containing high amounts of carbohydrates and CN [9,14,39,45].

CN degradation in cassava waste

Over the years, physicochemical and biotechnological technologies have been used, and improved, to remove dangerous contaminants such as CN [148]. Besides cassava manufacturing, other industries such as pesticides, cosmetics, electroplating, mining activities, galvanizing, and jewelry, contribute to the increase of CN in the environment [18,22,123,149,150]. In addition, anthropogenic forms of CN^- and HCN can enter the environment in several ways, it can be deposited or leached into the soil and

Influences of CN on the degradation of other pollutants and VAP generation

There is a gap in understanding how CN can interfere in cassava treatment and obtain by-products from these wastes. Many studies have not analyzed this influence, but considering how CN can be harmful or beneficial to treatment, and microorganisms must be addressed [62,65,69,195]. However, the number of CN-degrading microorganisms is considerably high; the insight for this analysis in cassava waste is rare, as pretreatment can remove CN. According to Dash et al. [126], the sensitivity of

Trends in CN biodegradation from cassava waste

Conventional physicochemical methods are ineffective in removing high concentrations of CN in industrial wastewater. In addition, they present other disadvantages such as high costs, not being eco-friendly, maintenance difficulty, and toxic compounds production [21,149,150,156,168,[200], [201], [202]]. On the other hand, CN removal from industrial wastewater can be carried out by microorganisms that can degrade it [22]. Therefore, the biological models are an interesting approach once CN can be

Conclusions

In this review, the CN removal in cassava waste through bioprocesses was reported, also discussed how to embrace the circular economy. It was demonstrated that the cassava industry generates significant amounts of liquid and solid effluents whose composition varies according to the process, mainly for CN. Anaerobic and aerobic biological processes are efficient in removing CN from these wastes. Therefore, herein, it was discussed that biological processes can biodegrade high amounts of CN, but

CRedit authorship contribution statement

Hérick Claudino Mendes: Writing – original draft, Methodology, Investigation, Data curation, Conceptualization. **Maikon Kelbert:** Writing – review & editing, Supervision. **Camila Michels:** Writing – review & editing, Supervision. **Hugo Moreira Soares:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The present study was performed in the framework of research work at the Federal University of Santa Catarina in Brazil and was financially supported by CAPES (Coordination for the Improvement of Higher-Level Personnel) (Process number 88887.511709/2020-00) for a scholarship.